「Prove that if $\{a_n\}$ is sequence of integers s.t infinitely many non-zero terms then the convergence bound of $\sum a_n$ is at most 1.

Sol). If $a_n = 1$ $\forall n \in \mathbb{N}$, then $R = 1$.

And, by the Theorem $R > 1 \Leftrightarrow \limsup\limits_{n \to \infty} \sqrt[n]{|a_n|} < 1$.

Put $\limsup \sqrt[n]{a_n} = \beta < 1$. Then, for $\beta < \alpha < 1$,

$$\exists N \in \mathbb{N} \text{ s.t } n \geq N \Rightarrow \sqrt[n]{|a_n|} < \alpha$$

$$\Leftrightarrow |a_n| < \alpha^n < 1$$

Contradiction.

「Section 3, 11.

Let $a_n > 0$, $S_n = a_1 + \cdots + a_n$, and $\sum a_n$ diverges. Prove that

a) $\sum \frac{a_n}{1+a_n}$ diverges.

b) $\sum \frac{a_n}{S_n}$ diverges.

$a_n \le \frac{1}{2}(1+a_n)$

c) $\sum \frac{a_n}{S_n^2}$ converges.

$\Rightarrow a_n \le 1$

d) $\sum \frac{a_n}{1+na_n}$ and $\sum \frac{a_n}{1+n^2 a_n}$ Converges?

Sol a) Suppose that $\{a_n\}$ is bounded by $M > 0$. Then, $a_n < M$ implies $\frac{1}{1+a_n} > \frac{1}{1+M}$.

Therefore $\displaystyle\sum_{n=1}^{m} \frac{a_n}{1+a_n} > \sum_{n=1}^{m} \frac{a_n}{1+M} = \frac{1}{1+M} \cdot \sum_{n=1}^{m} a_n$

diverges.

And, if $\{a_n\}$ is not bounded, then there are infinitely many $n \in \mathbb{N}$ s.t $a_n > 1$.

This implies that $\frac{a_n}{1+a_n} > \frac{a_n}{a_n + a_n} = \frac{1}{2}$ for infinitely $n \in \mathbb{N}$. Therefore diverges.

Alternatively, using contraposition: Suppose $\sum \frac{a_n}{1+a_n}$ converges. Take $N \in \mathbb{N}$ s.t

$$n \ge N \Rightarrow \frac{a_n}{1+a_n} < \frac{1}{2}$$

It equals to $a_n < 1$. Now, for $n \ge N$, $\frac{a_n}{2} \le \frac{a_n}{1+a_n}$ implies $\sum \frac{a_n}{2}$ converges.

(Thanks 범희!)

Sol b). For any $N \in \mathbb{N}$ and $k \ge 0$,

$$\frac{a_{N+1}}{S_{N+1}} + \cdots + \frac{a_{N+k}}{S_{N+k}} \ge \frac{1}{S_{N+k}} \cdot (a_{N+1} + \cdots + a_{N+k}) = \frac{S_{N+k} - S_N}{S_{N+k}} = 1 - \frac{S_N}{S_{N+k}} \quad \cdots (*)$$

If $\sum \frac{a_n}{S_n}$ converges, then there exists $N \in \mathbb{N}$ s.t

$$m \ge n \ge N \Rightarrow \left| \sum_{k=n}^{m} \frac{a_n}{S_n} \right| < \frac{1}{2}.$$

But, since the inequality $(*)$, and $S_n = \sum_{i=0}^{n} a_i$ diverges, we can choose $k \ge N$ s.t

$$\frac{1}{2} \le 1 - \frac{S_N}{S_{N+k}} \le \sum_{i=N+k}^{N+k} \frac{a_i}{S_i}$$

which is contradiction.

Sol c) Since $s_{n-1} < s_n$ for any $n \in \mathbb{N}$,

$$\frac{a_n}{s_n^2} \le \frac{a_n}{s_{n-1}s_n} = \frac{s_n - s_{n-1}}{s_{n-1}\cdot s_n} = \frac{1}{s_{n-1}} - \frac{1}{s_n} \qquad (n \ge 2)$$

Now, for any $n \ge 2$,

$$\sum_{k=2}^{n} \frac{a_k}{s_k^2} \le \sum_{k=2}^{n} \frac{1}{s_{k-1}} - \frac{1}{s_k} = \frac{1}{s_1} - \frac{1}{s_n} \to \frac{1}{s_1} \quad \text{as } n \to \infty$$

Since s_n diverges to ∞.

Sol d) First,

$$\frac{a_n}{1+n^2a_n} \le \frac{a_n}{n^2a_n} = \frac{1}{n^2}, \quad \text{therefore } \sum \frac{a_n}{1+n^2a_n} \le \sum \frac{1}{n^2}, \text{ converges.}$$

To show the second assertion,
if $\{na_n\}$ is bounded by $M > 0$, then

$$na_n < M \text{ implies } \frac{a_n}{1+na_n} > \frac{a_n}{1+M}, \quad \text{therefore } \sum \frac{a_n}{1+na_n} > \frac{1}{1+M} \cdot \sum a_n \text{ diverges.}$$

if $\{na_n\}$ is <u>Not</u> bounded, then we can not determine Convergence. For instance, define

$$a_n = \begin{cases} 1 & \text{if } n = k^2 \text{ for some } k \in \mathbb{N} \\ \frac{1}{n^2} & \text{otherwise} \end{cases}$$

Then, $\sum a_n$ diverges since a_n does not converges to 0,
but $\sum \frac{a_n}{1+na_n}$ Converges, because

$$\frac{a_n}{1+na_n} = \begin{cases} \frac{1}{1+n} & \text{if } n = k^2 \text{ for some } k \in \mathbb{N} & \left(< \frac{1}{n}\right) \\ \frac{1}{n^2+n} & \text{otherwise} & \left(< \frac{1}{n^2}\right) \end{cases}$$

which satisfies $\displaystyle\sum_{n=1}^{m} \frac{a_n}{1+na_n} \le \sum_{n=1}^{m} \frac{1}{n^2} < \infty$ for all $m \in \mathbb{N}$.

Meanwhile, put $a_n = \frac{1}{n}$. Then both

$$\sum a_n, \quad \sum \frac{a_n}{1+na_n} = \sum \frac{1}{2n}$$

are divergent.

Section 3, 12

Let $a_n > 0$, $\sum a_n$ converges, put $r_n = \sum_{m=n}^{\infty} a_m$

a). Prove $\forall N, k \in \mathbb{N}$,

$$\frac{a_N}{r_N} + \cdots + \frac{a_{N+k}}{r_{N+k}} > 1 - \frac{r_{N+k}}{r_N}$$

Deduce that $\sum \frac{a_n}{r_n}$ diverges.

$$\frac{r_N - a_N}{r_N} = \frac{r_{N+1}}{r_N}$$

b). Prove $\forall n \in \mathbb{N}$,

$$\frac{a_n}{\sqrt{r_n}} < 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

Deduce that $\sum \frac{a_n}{\sqrt{r_n}}$ Converges.

Sol). Since $\sum a_n$ Converges, put $A = \sum_{n=1}^{\infty} a_n$. Then,

$r_n \to 0$?

$$A - r_n = \sum_{k=1}^{n-1} a_k$$

And, r_n is decreasing. Now, $r_N > r_{N+i} \Rightarrow \frac{1}{r_N} < \frac{1}{r_{N+i}}$ implies

$$\frac{a_N}{r_N} + \cdots + \frac{a_{N+k}}{r_{N+k}} > \frac{1}{r_N}(a_N + a_{N+1} + \cdots + a_{N+k})$$

$$= \frac{1}{r_N}(r_N - r_{N+k+1})$$

$$= 1 - \frac{r_{N+k+1}}{r_N} > 1 - \frac{r_{N+k}}{r_N}$$

Now, suppose $\sum \frac{a_n}{r_n}$ Converges. Then, there exists $N \in \mathbb{N}$ s.t for all $k \in \mathbb{N}$,

$$\left| \sum_{n=N}^{N+k} \frac{a_n}{r_n} \right| < \frac{1}{2}.$$

Meanwhile, $\lim_{n \to \infty} r_n = 0$ because for any $\varepsilon > 0$, We can choose $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow |r_n| = r_n = \sum_{m=n}^{\infty} r_m = A - \sum_{k=1}^{n-1} a_k < \varepsilon$$

Therefore, We can choose $K \in \mathbb{N}$ for fixed N,

$$k \geq K \Rightarrow \left| \sum_{n=N}^{N+k} \frac{a_n}{r_n} \right| > 1 - \frac{r_{N+k}}{r_N} \geq \frac{1}{2}.$$

It is Contradiction.

sol b). since $a_n = r_n - r_{n+1} = (\sqrt{r_n} - \sqrt{r_{n+1}})(\sqrt{r_n} + \sqrt{r_{n+1}})$ $(n \geq 1)$

$$\frac{a_n}{\sqrt{r_n}} = (\sqrt{r_n} - \sqrt{r_{n+1}})\left(1 + \frac{\sqrt{r_{n+1}}}{\sqrt{r_n}}\right)$$

$$< (\sqrt{r_n} - \sqrt{r_{n+1}})(1+1) = 2(\sqrt{r_n} - \sqrt{r_{n+1}})$$

Therefore,

$$\sum_{k=1}^{n} \frac{a_k}{\sqrt{r_k}} < \sum_{k=1}^{n} 2(\sqrt{r_k} - \sqrt{r_{k+1}})$$

$$= 2(\sqrt{r_1} - \sqrt{r_{n+1}}) \to 2\sqrt{r_1} \quad \text{as } n \to \infty.$$

Thus $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

「Section 3.13

Given absol. convergent $\sum a_n$, $\sum b_n$, $\sum_{r=n}^{\infty}\sum_{k=1}^{n} a_{n-k} b_k$ converges absolutely.

Prof. Denote $A_n = \sum_{k=0}^{n} |a_k|$, $B_n = \sum_{k=0}^{n} |b_k|$, and $C_n = \sum_{k=0}^{n} \left| \sum_{r=0}^{k} a_{k-r} \cdot b_r \right|$ $\quad \beta_n = B - B_n$

Put $A = \lim_{n\to\infty} A_n$, $B = \lim_{n\to\infty} B_n$

$$C_n = |a_0 b_0| + |a_1 b_0 + a_0 b_1| + |a_2 b_0 + a_1 b_1 + a_0 b_2| + \cdots + |a_n b_0 + \cdots + a_0 b_n|$$

$$\leq |a_0| B_n + |a_1| B_{n-1} + \cdots + |a_n| B_0$$

$$= |a_0|(B - \beta_n) + |a_1|(B - \beta_{n-1}) + \cdots + |a_n|(B - \beta_0)$$

$$= A_n B - (|a_0|\beta_n + |a_1|\beta_{n-1} + \cdots + |a_n|\beta_0)$$

$$\leq A_n B \to AB$$

Therefore, C_n is monotone increasing bounded sequence, it is converges.

Section 3.14.

Let $\{S_n\}$ be a complex numbers sequence, and define

$$\sigma_m = \frac{S_1 + S_2 + \cdots + S_m}{m} \quad (m \in \mathbb{N})$$

$(m\sigma_m - n\sigma_n)$

a. If $\lim_{n\to\infty} S_n = S$, then $\lim_{n\to\infty} \sigma_n = S$

sol) Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ st

$$n \geq N \Rightarrow |S_n - S| < \frac{\varepsilon}{2}$$

meanwhile, observe that for $m > N$,

$$|m\sigma_m - mS| = |(S_1 - S) + \cdots + (S_N - S) + (S_{N+1} - S) + \cdots + (S_m - S)|$$

$$\leq |(S_1 - S) + \cdots + (S_N - S)| + |S_{N+1} - S| + \cdots + |S_m - S|$$

$$\leq |(S_1 - S) + \cdots + (S_N - S)| + m \cdot \frac{\varepsilon}{2} \quad (m\tfrac{\varepsilon}{2} \text{ can be replaced by } (m-N+1)\tfrac{\varepsilon}{2})$$

Taking $\frac{1}{m}$ both side, then

$$|\sigma_m - S| \leq \frac{1}{m} \cdot |(S_1 - S) + \cdots + (S_N - S)| + \frac{\varepsilon}{2}$$

Now, Take $M \in \mathbb{N}$ st $|(S_1 - S) + \cdots + (S_N - S)| < M\frac{\varepsilon}{2}$, then

$$m \geq M \Rightarrow |\sigma_m - S| < \varepsilon.$$

a-1) If $\{S_n\}$ is a real numbers, then

$$\liminf_{n\to\infty} S_n \leq \liminf_{n\to\infty} \sigma_n \leq \limsup_{n\to\infty} \sigma_n \leq \limsup_{n\to\infty} S_n$$

proof. Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ st

$$n \geq N \Rightarrow S_n < \limsup_{n\to\infty} S_n + \frac{\varepsilon}{2}$$

$$\sigma_n = \frac{S_1 + \cdots + S_N + S_{N+1} + \cdots + S_n}{n} = \frac{1}{n}(S_1 + \cdots + S_N) + \frac{1}{n}(S_{N+1} + \cdots + S_n)$$

$$\leq \frac{1}{n}(S_1 + \cdots + S_N) + \frac{1}{n} \cdot (n-N) \cdot \left(\limsup_{n\to\infty} S_n + \frac{\varepsilon}{2}\right)$$

$$\leq \frac{1}{n}(S_1 + \cdots + S_N) + \limsup S_n + \frac{\varepsilon}{2}$$

Take $M \in \mathbb{N}$ st $(S_1 + \cdots + S_N) < M \cdot \frac{\varepsilon}{2}$. Then, for $n \geq M$,

$$\sigma_n \leq \limsup_{n\to\infty} S_n + \varepsilon.$$

Since $\varepsilon > 0$ was chosen arbitrarily, $\limsup_{n\to\infty} \sigma_n \leq \limsup_{n\to\infty} S_n$.

d). Put $a_n = s_{n+1} - s_n$; $n \ge 1$. Show that

$$s_{n+1} - \sigma_n = \frac{1}{n} \cdot \sum_{k=1}^{n} k a_k$$

And deduce that $\lim_{n\to\infty} n a_n = 0$ and $\lim_{n\to\infty} \sigma_n = s$ implies $\lim_{n\to\infty} s_n = s$.

proof. Since

$$\begin{aligned}\sum_{k=1}^{n} k a_k &= a_1 + 2a_2 + 3a_3 + \cdots + n a_n \\ &= (s_2 - s_1) + 2(s_3 - s_2) + \cdots + n(s_{n+1} - s_n) \\ &= n s_{n+1} - s_n - s_{n-1} - \cdots - s_1 \\ &= n s_{n+1} - (s_1 + \cdots + s_n) \\ &= n s_{n+1} - n \cdot \sigma_n\end{aligned}$$

Therefore,

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \Big(\sigma_n + \underbrace{\frac{1}{n} \cdot \sum_{k=1}^{n} k a_k}_{=0,\ \text{by a)}} \Big) = \lim_{n\to\infty} \sigma_n = s.$$

e) Assume $\{n a_n\}$ bounded and $\lim_{n\to\infty} \sigma_n = s$, then $\lim_{n\to\infty} s_n = s$.

Proof. Let $M > 0$ be a bound of $\{n a_n\}$.

Observe that for $n > m$,

($n > m$, then $\frac{1}{n} = \frac{m}{n-m}\left(\frac{1}{m} - \frac{1}{n}\right)$, it is derived by $\frac{1}{AB} = \frac{1}{B-A}\left(\frac{1}{A} - \frac{1}{B}\right)$)

$$\begin{aligned}s_n - \sigma_n &= s_n - \frac{1}{n} \cdot (s_1 + \cdots + s_n) \\ &= s_n - \frac{m}{n-m}\left(\frac{1}{m} - \frac{1}{n}\right)(s_1 + \cdots + s_n) \\ &= s_n - \frac{m}{n-m}\left(\frac{1}{m}(s_1 + \cdots + s_m + s_{m+1} + \cdots + s_n) - \frac{1}{n}(s_1 + \cdots + s_n)\right) \\ &= s_n - \frac{m}{n-m}\left(\frac{1}{m}\left[m\sigma_m + (s_{m+1} + \cdots + s_n)\right] - \sigma_n\right) \\ &= s_n - \frac{m}{n-m}\left(\sigma_m - \sigma_n + \frac{1}{m} \cdot \left[s_{m+1} + \cdots + s_n\right]\right) \\ &= \frac{m}{n-m}(\sigma_n - \sigma_m) + \frac{1}{n-m} \cdot \sum_{k=m+1}^{n} (s_n - s_k)\end{aligned}$$

$-1 + m\varepsilon \le n \iff \frac{m}{n-m+1} \le \frac{1}{\varepsilon}$

$n - \varepsilon < m + m\varepsilon \iff \frac{n-m}{-} < \varepsilon$

Meanwhile, since $|n a_n| = n|s_{n+1} - s_n| < M$, for $i \ge m+1$,

$$\begin{aligned}|s_n - s_i| &= |s_n - s_{n-1} + s_{n-1} - s_{n-2} + s_{n-2} - \cdots + s_{i+1} - s_i| \\ &\le |s_n - s_{n-1}| + |s_{n-1} - s_{n-2}| + \cdots + |s_{i+1} - s_i| \\ &\le \frac{M}{n-1} + \frac{M}{n-2} + \cdots + \frac{M}{i} \le \frac{n-i}{i} \cdot M \le \frac{n-i}{m+1} \cdot M \le \frac{n-m-1}{m+1} \cdot M.\end{aligned}$$

Given $\varepsilon > 0$ and for each $n \in \mathbb{N}$, take $m \in \mathbb{N}$ satisfying $m \overset{(1)}{\le} \frac{n-\varepsilon}{1+\varepsilon} \overset{(2)}{<} m+1$

Then, $m < n$ and $(1) \iff m + m\varepsilon + \varepsilon \le n \iff \frac{m+1}{n-m} \le \frac{1}{\varepsilon}$

and $(2) \iff n - \varepsilon \le m + 1 + \varepsilon m + \varepsilon \iff \frac{n-m-1}{m+2} < \varepsilon$ implies if $i \ge m+1$, $|s_n - s_i| \le \frac{n-m-1}{m+1} \cdot M < \varepsilon M$.

Finally

$$\begin{aligned}\limsup_{n\to\infty} |s_n - s| &\le \limsup_{n\to\infty} |s_n - \sigma_n| + \limsup_{n\to\infty} |\sigma_n - s| = \limsup_{n\to\infty} |s_n - \sigma_n| \\ &\le \limsup_{n\to\infty} \left(\frac{m}{n-m}|\sigma_n - \sigma_m|\right) + \limsup_{n\to\infty} \left(\frac{1}{n-m} \cdot \sum_{k=m+1}^{n} |s_n - s_k|\right) \\ &\le \frac{1}{\varepsilon} \limsup_{n\to\infty} |\sigma_n - \sigma_m| + \varepsilon M = \varepsilon M\end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, $\lim_{n\to\infty} s_n = s$.

Section 3.16.

Fix $\alpha > 0$, and choose $x_1 > \sqrt{\alpha}$. Define $\{x_n\}$ as recursively:

$$x_{n+1} := \frac{1}{2} \cdot \left(x_n + \frac{\alpha}{x_n} \right)$$

a). $\{x_n\}$ decreases monotonically, and $\lim_{n\to\infty} x_n = \sqrt{\alpha}$

Proof. observe that

$$x_{n+1} \leq x_n \iff \alpha \leq x_n^2 \iff \frac{\alpha}{x_n} \leq x_n$$

For $n=1$, it is trivially true. Suppose for $n \in \mathbb{N}$, it is true. Then,

$$x_{n+1} = \frac{1}{2} \cdot \left(x_n + \frac{\alpha}{x_n} \right) \leq \frac{1}{2} \cdot \left(x_n + x_n \right) = x_n.$$

Point; $\frac{\alpha}{x_n} \leq x_n$

By induction, $\{x_n\}$ is monotone decreasing. And $x_n \geq \sqrt{\alpha}$ for all $n \in \mathbb{N}$,

$\{x_n\}$ is decreasing bounded, therefore converges. Put $x = \lim_{n\to\infty} x_n \geq \sqrt{\alpha}$. Now,

$$x = \lim_{n\to\infty} x_{n+1} = \frac{1}{2} \left(\lim_{n\to\infty} x_n + \frac{\alpha}{\lim_{n\to\infty} x_n} \right) = \frac{1}{2} \left(x + \frac{\alpha}{x} \right) \Rightarrow x = \sqrt{\alpha}.$$

b) Put $\varepsilon_n = x_n - \sqrt{\alpha}$ (That is, error between x_n and $\sqrt{\alpha}$, magnitude for approximation)

Show that $\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$ and that $\varepsilon_{n+1} < 2\sqrt{\alpha} \cdot \left(\frac{\varepsilon_1}{2\sqrt{\alpha}} \right)^{2^n}$ for $n \in \mathbb{N}$

Proof since

$$\varepsilon_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} = \frac{1}{2} \cdot \frac{\varepsilon_n^2 + 2\varepsilon_n\sqrt{\alpha} + 2\alpha}{x_n} - \sqrt{\alpha} = \frac{\varepsilon_n^2}{2x_n} + \frac{\sqrt{\alpha}}{x_n} \cdot (\varepsilon_n + \sqrt{\alpha}) - \sqrt{\alpha} = \frac{\varepsilon_n^2}{2x_n}$$

and $x_n \geq \sqrt{\alpha}$ gives $\varepsilon_{n+1} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$. Now, inductively,

$$\varepsilon_{n+1} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}} < \frac{1}{2\sqrt{\alpha}} \cdot \left(\frac{\varepsilon_{n-1}^4}{4\alpha} \right) = \frac{1}{2\sqrt{\alpha}} \cdot \frac{1}{(2\sqrt{\alpha})^2} \cdot \varepsilon_{n-1}^4 < \cdots \quad \left(\sum_{k=0}^{n-1} 2^k = 2^n - 1 \right)$$

$$< \left(\frac{1}{2\sqrt{\alpha}} \right)^{1+2+2^2+\cdots+2^{n-1}} \cdot \varepsilon_{n-(n-1)}^{2^n} = 2\sqrt{\alpha} \cdot \left(\frac{\varepsilon_1}{2\sqrt{\alpha}} \right)^{2^n}$$

17. Fix $\alpha > 1$, take $x_1 > \sqrt{\alpha}$. Define recursively

$$x_{n+1} = \frac{\alpha + x_n}{1 + x_n} = x_n + \frac{\alpha - x_n^2}{1 + x_n}$$

a, b) Prove $x_1 > x_3 > x_5 > \cdots$ and $x_2 < x_4 < x_6 < \cdots$

Proof. Observe that for $n \in \mathbb{N}$, $x_{n+1} = \frac{\alpha + x_n}{1 + x_n} > \frac{1 + x_n}{1 + x_n} = 1$. And

$$x_{n+1}^2 = \frac{\alpha^2 + 2\alpha x_n + x_n^2}{1 + 2x_n + x_n^2} = \frac{\alpha(1 + 2x_n + x_n^2) + \alpha^2 - \alpha - (\alpha - 1)x_n^2}{1 + 2x_n + x_n^2} = \alpha + \underbrace{(\alpha-1)}_{\geq 0} \cdot \frac{\alpha - x_n^2}{1 + 2x_n + x_n^2}$$

Therefore, $\begin{cases} x_{n+1} < \sqrt{\alpha} & \text{if } x_n > \sqrt{\alpha} \\ x_{n+1} > \sqrt{\alpha} & \text{if } x_n < \sqrt{\alpha}. \end{cases}$ implies $\begin{cases} x_{2n} < \sqrt{\alpha} \\ x_{2n-1} > \sqrt{\alpha} \end{cases}$ for all $n \in \mathbb{N}$.

Meanwhile,

$$x_{n+2} = \frac{\alpha + x_{n+1}}{1 + x_{n+1}} = \frac{\alpha + \frac{\alpha + x_n}{1 + x_n}}{1 + \frac{\alpha + x_n}{1 + x_n}} = \frac{2\alpha + (\alpha+1)x_n}{2x_n + \alpha + 1} = x_n \cdot \frac{2 \cdot \frac{\alpha}{x_n} + \alpha + 1}{2x_n + \alpha + 1}$$

For $n = 2k$, $x_{2k} < \sqrt{\alpha}$ implies $x_{2k} < \frac{\alpha}{x_{2k}}$ and

$n = 2k-1$, $x_{2k-1} > \sqrt{\alpha}$ implies $x_{2k-1} > \frac{\alpha}{x_{2k-1}}$

these gives; $x_{2k+2} > x_{2k}$ and $x_{2k+1} < x_{2k-1}$ for all $k \in \mathbb{N}$. ∎

c) $\lim_{n \to \infty} x_n = \sqrt{\alpha}$

proof. Since $\{x_{2n}\}$ and $\{x_{2n-1}\}$ both are increasing and decreasing, respectively, bounded, Therefore both are convergent. Put $L_1 = \lim_{n\to\infty} x_{2n+1}$ and $L_2 = \lim_{n\to\infty} x_{2n}$. Then,

$$L_1 = \lim_{n\to\infty} x_{2n+1} = \frac{\alpha + \lim_{n\to\infty} x_{2n}}{1 + \lim_{n\to\infty} x_{2n}} = \frac{\alpha + L_2}{1 + L_2} \text{ and similarly } L_2 = \frac{\alpha + L_1}{1 + L_1}$$

$$L_1 = \frac{\alpha + \frac{\alpha + L_1}{1 + L_1}}{1 + \frac{\alpha + L_1}{1 + L_1}} = \frac{\alpha + \alpha L_1 + \alpha + L_1}{1 + L_1 + \alpha + L_1} = \frac{2\alpha + (\alpha+1)L_1}{2L_1 + (\alpha+1)} \Rightarrow L_1 = \sqrt{\alpha}.$$

Similarly $L_2 = \sqrt{\alpha}$. Finally,

$$\sqrt{\alpha} \leq \liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n \leq \sqrt{\alpha}$$

implies $\lim_{n\to\infty} x_n = \sqrt{\alpha}$

Similarly, define $\varepsilon_n = |x_n - \sqrt{\alpha}|$.. Then,

$$\varepsilon_{n+1} = |x_{n+1} - \sqrt{\alpha}| = \left| \frac{\alpha + x_n}{1 + x_n} - \sqrt{\alpha} \right|$$

Similarly, define $\varepsilon_n = |x_n - \sqrt{\alpha}|$. Then, $\varepsilon_n = \begin{cases} x_n - \sqrt{\alpha} & \text{if } n \text{ odd} \\ \sqrt{\alpha} - x_n & \text{if } n \text{ even} \end{cases}$

If n is even, then

$$\varepsilon_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{\alpha + x_n}{1 + x_n} = \frac{\alpha + \sqrt{\alpha} - \varepsilon_n}{1 + \sqrt{\alpha} - \varepsilon_n} - \sqrt{\alpha} = \frac{(\sqrt{\alpha} - 1) \cdot \varepsilon_n}{1 + \sqrt{\alpha} - \varepsilon_n}$$

$$= \frac{}{1 + \sqrt{\alpha} - \varepsilon_n}$$

⌜Section 3.20.

Suppose $\{p_n\}$ is a Cauchy sequence in a Metric Space (X, d).

If some subsequence $\{p_{n_i}\}$ converges to p, then $\{p_n\}$ converges to p.

Proof. Given $\varepsilon > 0$, take $N_1 \in \mathbb{N}$ s.t.

$$m, n \geq N_1 \Rightarrow d(p_m, p_n) < \frac{\varepsilon}{2}$$

and take $N_2 \in \mathbb{N}$ s.t.

$$k \geq N_2 \Rightarrow d(p_{n_k}, p) < \frac{\varepsilon}{2}$$

Then, for $r \geq N = \max\{N_1, N_2\}$,

$$d(p_r, p) \leq d(p_r, p_{n_N}) + d(p_{n_N}, p) < \varepsilon.$$

($n_N \geq N$ by definition)

⌜Section 3.21.

Let (X, d) be a Complete Metric space, and $\{E_n\}$ be a non-empty closed bounded subset of X s.t.

$$E_1 \supseteq E_2 \supseteq \cdots$$

If $\lim_{n\to\infty} \operatorname{diam} E_n = 0$, then $\bigcap_{n=1}^{\infty} E_n$ is singleton.

Proof. If $E = \bigcap_{n=1}^{\infty} E_n$ has two points, $x, y \in E$, then

$$d(x,y) \leq \lim_{n\to\infty} \operatorname{diam} E_n = 0$$

Which is contradiction. Thus, enough to show that E is not empty set.

For each $k \in \mathbb{N}$, $\bigcap_{i=1}^{k} E_i = E_k$ is not empty set, by assumption, therefore we can choose $x_k \in E_k$.

Then, the sequence $\{x_k\}$ is a Cauchy Sequence, because; let $\varepsilon > 0$ be given.

Take $N \in \mathbb{N}$ s.t. $\operatorname{diam} E_N < \varepsilon$. Then, for any $m, n \geq N$,

$$d(x_m, x_n) \leq \operatorname{diam} E_N < \varepsilon.$$

Therefore, $\{x_k\}$ is Cauchy sequence in X. Since X is complete, put $x = \lim_{n\to\infty} x_n$.

Clearly, for any $n \in \mathbb{N}$, $\{x_k\}_{k=n}^{\infty} \subseteq E_n = \overline{E_n}$, therefore $x \in E_n$ for all $n \in \mathbb{N}$.

Finally, $x \in E$.